

# Oluwatobi Melvyn Mayungbo

Data and Machine Learning Analyst

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## SUMMARY

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Data and machine learning analyst with a B.Sc. in Statistics. Finds the real driver behind a problem and builds the fix, including detection and scoring services, LLM pipelines with output checks, and experiments reported honestly, even when a result did not work. Works in SQL and Python.

## SKILLS

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**Data Analysis:** SQL, Pandas, Anomaly Detection, NumPy

**Statistics and Causal Inference:** Experiment Design, Power Analysis, CUPED, Statistical Inference, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

**AI and LLM Systems:** Prompt Engineering, LLM Evaluation and Guardrails, Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

**Engineering and MLOps:** Python, FastAPI, SQL, DuckDB, PostgreSQL, Redis, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

**Machine Learning:** PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Payments and Risk:** Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

## PROJECTS

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### Uplift Modelling for Campaign Targeting | [GitHub](#)

*Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions*

- Ran a causal inference study on the Hillstrom randomised e-mail trial (64,000 customers) to decide which customers to contact under a fixed budget, delivered as a two-page decision memo for a marketing director.
- Showed the campaign works (1.25% conversion against 0.57%) but that a non-randomised list overstates the effect by 7.5×, and that a response model mails five times as many customers the campaign harms as an uplift model does.
- Evaluated with Qini curves, AUUC, decile tables and bootstrap intervals; uplift targeting was worth about \$1,390 against about \$3,170 for raising the budget, and a failed placebo test on the primary campaign was reported openly.

### Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

*Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL*

- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Engineered a multi-stage reconciliation pipeline using DuckDB and FastAPI, with temporal join logic and tolerance thresholds to handle asynchronous processor webhooks and bank settlement files.
- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

### Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

*Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run*

- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never retrieved.
- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.

### Sequence-Based Transaction Fraud Detection System | [GitHub](#)

*Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP*

- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

## EXPERIENCE

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**Machine Learning Instructor, SQI College of ICT** March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

**Internal Auditor, NISER** May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

## EDUCATION

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**B.Sc. Statistics, University of Ibadan** January 2021 to March 2025

**Professional Diploma in Artificial Intelligence, SQI College of ICT** July 2025 to Present

## CERTIFICATIONS

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action